

A gap is a scale the classical theory lacks: solved low-dimensional mass gaps and the positive-action question

The Yang–Mills mass gap and this programme’s positive action floor are the same kind of statement: the infimum of an observable over nontrivial states is a positive finite multiple of a unit formed from the theory’s fixed constants, and it is zero or infinite whenever those constants admit no such unit. Classical Yang–Mills in four dimensions carries no constant at all in its field equations, and classical mechanics with a mass and a force carries no action unit, so both classical floors are zero; the classical scale covariance that realizes zero is the same similarity in both cases. Every low-dimensional mass gap that can be solved in writing then falls into two classes. A *supplied* gap is proportional to a constant written into the definition: the box gap $2\pi\hbar c/L$ of a free field, the strong-coupling lattice gap in units of $1/a$, a mass term, the string tension g^2L of two-dimensional Yang–Mills on a circle, and the repository’s own Ising and cluster gaps $Ka[1 - \tanh 2b]$ and J . A *generated* gap arises from the interplay of two structures that are each scale-free: in Yang–Mills quantum mechanics the commutator potential and the canonical commutator produce $\Delta = \delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$, which vanishes if either $\hbar$ or g is removed, and this is the exact form of “commutative fields have no gap” in zero space dimensions; in scale-invariant quantum mechanics a scale is forced by the consistency requirement of defining the dynamics at all. The dictionary in Section 7 records what this decides for the necessity theorem sought in STATE item 1: the mass-gap problem takes the non-commutative structure as given and asks for the scale, so the analogy governs the second stage of the positive-action question and identifies the consistency-forced scale as the only mechanism in the mass-gap story that does not assume the structure it explains. This is G07, an exploratory consolidation with written proofs; it promotes no ledger claim.

1. Two floors and one dimensional theorem

The mass gap. Jaffe and Witten require a self-adjoint $H \geq 0$ with $H\Omega = 0$, $\text{spec}(H) \cap (0, \Delta) = \emptyset$ for some $\Delta > 0$, and a finite supremum m of such Δ (definitions). For a purely discrete spectrum $E_0 < E_1 < \dots$ the supremum is attained and equals $E_1 - E_0$.

The action floor. Q14 defines, for a model with fixed constants c , the attained set $\mathcal{A}(c)$ of an action observable over admitted masses, preparations and solutions, and the floor $g(c) = \inf\{A \in \mathcal{A}(c) : A > 0\}$. The programme’s target is $g(c) > 0$ for the Galileo comparison, whose observable is $\tau\Delta E = (3F/v)A_{\text{inertial,fall}} = F^2\tau^3/(2m)$ (N01 §1).

Theorem 1 (floors are unit multiples). Let a theory be specified by fixed constants $c_1, \dots, c_r$ with dimension vectors $d_i \in \mathbb{R}^3$ over mass, length and time, and let X be an observable with dimension vector d_X whose attained set $\mathcal{X}(c) \subset [0, \infty]$ is defined by dimensionally homogeneous relations and whose floor $\gamma(c) = \inf(\mathcal{X}(c) \cap (0, \infty))$ depends on the fixed constants only. Then $\gamma(c) \in \{0, \infty\}$ for every c when d_X lies outside the span of $d_1, \dots, d_r$, and $\gamma(c) = \Pi(c)F(\pi_1, \dots, \pi_s)$ with Π a monomial of dimension d_X and π_j the independent dimensionless combinations otherwise.

Proof. This is Q14’s Theorem A with d_A replaced by d_X , and the proof is the same three lines. A change of units $\lambda \in (0, \infty)^3$ carries the model at c to the model at $\lambda^{d_i}c_i$ and multiplies every attained value by λ^{d_X} , so $\gamma(\lambda^{d_1}c_1, \dots) = \lambda^{d_X}\gamma(c)$. If d_X is outside the span, choose λ with $\lambda^{d_i} = 1$ for all i and $\lambda^{d_X} \neq 1$: then $\gamma(c) = \lambda^{d_X}\gamma(c)$, which forces $\gamma(c) \in \{0, \infty\}$. If $d_X = \sum a_i d_i$, then γ/Π is invariant under all unit changes and is therefore a function of the dimensionless combinations. $\square$

Corollary 2 (similarity forces zero). If the admitted class at fixed constants is preserved by a one-parameter family of transformations that multiplies X by every $s > 0$, then $\gamma = 0$ whenever some positive value is attained. This is Q14’s Theorem B for the observable X .

The two instances that motivate this note:

	Mass gap of pure Yang–Mills	Action floor of the Galileo comparison
Classical fixed constants d_X in their span?	none in $D_A^* F_A = 0$; c alone no: c gives no mass	m, F (or m, g) no: $(1, 2, -1) \notin \text{span}\{(1, 0, 0), (1, 1, -2)\}$
Classical similarity realizing zero	$A_s(x) = sA(sx)$: static energy $\mapsto sE$ in three space dimensions	$x \mapsto sx, t \mapsto s^{1/2}t$: $\tau \Delta E \mapsto s^{3/2} \tau \Delta E$
Classical floor	0	0 (C002, C027, C056–C058)
Quantum fixed constants	$\hbar, c$, dimensionless $g^2 \hbar$: still no mass unit in $d = 4$	$\hbar$: action unit present
Where the scale must come from	the regularization scale, by transmutation (Section 4)	the constant $\hbar$ itself, whose necessity is the open question

The classical Yang–Mills row is checked in Section 4; the Galileo row is checked in Section 7. The table already states the analogy at the level of Theorem 1: both classical theories have an empty span, both quantum theories must obtain the missing unit from somewhere, and the two “somewheres” differ.

2. The exact $\hbar$ -scaling of spectra for homogeneous potentials

Theorem 3. Let $V \geq 0$ be continuous on $\mathbb{R}^n$ and positively homogeneous of degree $k > -2$, $V(sx) = s^k V(x)$, and let $H = -\frac{\hbar^2}{2m} \Delta + \lambda V$ be the Friedrichs operator of its form on C_0^∞ . With the unitary dilation $(U_\ell \psi)(x) = \ell^{-n/2} \psi(x/\ell)$ and

$$\ell^{k+2} = \frac{\hbar^2}{m\lambda}, \quad \varepsilon = \frac{\hbar^2}{m\ell^2} = \hbar^{\frac{2k}{k+2}} \lambda^{\frac{2}{k+2}} m^{-\frac{k}{k+2}},$$

one has $U_\ell^{-1} H U_\ell = \varepsilon (-\frac{1}{2} \Delta + V)$. Every spectral datum of H , in particular any gap, is ε times the corresponding pure number of $-\frac{1}{2} \Delta + V$.

Proof. $\Delta(U_\ell \psi) = \ell^{-2} U_\ell(\Delta \psi)$ and $V U_\ell \psi = \ell^k U_\ell(V \psi)$ on C_0^∞ , which U_ℓ preserves; so $U_\ell^{-1} H U_\ell = \frac{\hbar^2}{m\ell^2} (-\frac{1}{2} \Delta) + \lambda \ell^k V$, and the choice of ℓ equates the two coefficients. The identity passes to the Friedrichs extensions because U_ℓ maps the form on the core to the transformed form. Unitary equivalence preserves spectra and multiplicities. $\square$

Proposition 4 (classical similarity). For the classical Hamiltonian $|p|^2/2m + \lambda V(x)$ with V homogeneous of degree k , the map $x \mapsto sx, t \mapsto s^{1-k/2} t$ carries solutions to solutions, multiplies energies by s^k and the action along a trajectory by $s^{1+k/2}$ (Landau–Lifshitz, *Mechanics* §10; verified against the equation of motion: $m\ddot{x}$ scales as $s/s^{2-k} = s^{k-1}$ and $\lambda \nabla V$ as s^{k-1}). Hence for $k > 0$ the range of the classical Hamiltonian is $[0, \infty)$: the classical energy floor is zero, and the classical action floor is zero by Corollary 2.

Three readings of Theorem 3 organize the rest of the note.

- For $k > 0$ the exponent $2k/(k+2)$ is positive: the gap of H is $\varepsilon \delta$ with δ a pure number and vanishes as $\hbar \rightarrow 0$ and as $\lambda \rightarrow 0$. A gap of a growing potential is a quantum effect and an interaction effect at once; at $\lambda = 0$ the operator is $-\hbar^2 \Delta/2m$ with spectrum $[0, \infty)$.
- The degree $k = -2$ is action-critical: the classical action is scale-invariant, $m\lambda$ has the dimension of action squared, the exponent of ε has a pole, and the quantum problem depends on the single dimensionless number $2m\lambda/\hbar^2$ with no energy unit at all. This is the mechanical counterpart of the Yang–Mills scaling $S(A_s) = s^{4-d} S(A)$, which is critical at $d = 4$ (comparison §3). Section 6 treats it.
- Special cases of ε : $k = 2$ gives $\hbar \sqrt{\lambda/m}$, the oscillator quantum up to a pure number; $k = 4$ gives $\hbar^{4/3} \lambda^{1/3} m^{-2/3}$; $k \rightarrow \infty$ gives $\hbar^2/(m\ell^2)$ with ℓ the box length, the particle-in-a-box

unit; $k = -1$ gives $m\lambda^2/\hbar^2$, the Rydberg unit, whose *inverse* dependence on $\hbar$ means that the hydrogen level spacing diverges as $\hbar \rightarrow 0$ while the levels themselves crowd to zero from below. In each case the gap is the unique energy monomial in $(\hbar, m, \lambda)$, as Theorem 1 requires.

3. Yang–Mills quantum mechanics: a solved gap that needs two commutators

The zero-momentum sector of Yang–Mills theory on a torus is a finite-dimensional quantum-mechanical system whose potential is the squared commutator of the constant gauge potentials. Its spectrum is purely discrete for every positive $\hbar$ and coupling, although the classical energy range is $[0, \infty)$ (Simon 1983; Lüscher 1983). The proof below is Simon’s first argument with every constant explicit, extended to the $SU(2)$ model with D matrices and to its gauge-invariant sector.

3.1 The scalar model

Theorem 5. Let $H = -\frac{\hbar^2}{2m}\Delta + \frac{g^2}{2}x^2y^2$ on $L^2(\mathbb{R}^2)$, defined as the Friedrichs operator of its form on $C_0^\infty(\mathbb{R}^2)$.

1. As quadratic forms,

$$H \geq \frac{1}{2}K, \quad K = -\frac{\hbar^2}{2m}\Delta + \frac{\hbar g}{2\sqrt{m}}(|x| + |y|).$$

2. H has compact resolvent: its spectrum is a sequence $E_0 < E_1 < E_2 < \dots$ of eigenvalues of finite multiplicity tending to infinity.
3. E_0 is simple with a strictly positive eigenfunction, so $\Delta = E_1 - E_0 > 0$, and $H - E_0$ satisfies the Jaffe–Witten conditions with $0 < m = \Delta < \infty$.
4. $E_n(\hbar, m, g) = \varepsilon e_n$ with e_n the eigenvalues of $\frac{1}{2}(-\Delta + x^2y^2)$ and

$$\varepsilon = \hbar^{4/3}g^{2/3}m^{-2/3}, \quad \Delta = \delta_1 \hbar^{4/3}g^{2/3}m^{-2/3}, \quad \delta_1 = e_1 - e_0 > 0.$$

Hence $\Delta \rightarrow 0$ as $\hbar \rightarrow 0$ and as $g \rightarrow 0$. At $g = 0$ the spectrum is $[0, \infty)$ with no eigenvalue. The classical Hamiltonian $|p|^2/2m + g^2x^2y^2/2$ has range $[0, \infty)$, and the motions $y = p_y = 0$, $x(t) = x_0 + p_xt/m$ escape to infinity along the valley with arbitrarily small energy.

Proof. Oscillator bound. For $\phi \in C_0^\infty(\mathbb{R})$ and $\mu, \nu > 0$, with $\alpha = \sqrt{\nu/\mu}$, integration by parts gives $\int |\phi' + \alpha y \phi|^2 dy = \int |\phi'|^2 + \alpha^2 \int y^2 |\phi|^2 - \alpha \int |\phi|^2$, so

$$\int (\mu |\phi'|^2 + \nu y^2 |\phi|^2) dy \geq \sqrt{\mu\nu} \int |\phi|^2 dy.$$

With $\mu = \hbar^2/2m$ and $\nu = \frac{1}{2}m\omega^2$ this is the zero-point energy $\hbar\omega/2$ of the oscillator $-\frac{\hbar^2}{2m}\partial_y^2 + \frac{1}{2}m\omega^2y^2$.

Slicing. Fix x and apply the bound to $\phi = \psi(x, \cdot)$ with $\nu = g^2x^2/2$, that is $m\omega_x^2 = g^2x^2$, $\omega_x = g|x|/\sqrt{m}$, $\sqrt{\mu\nu} = \hbar g|x|/(2\sqrt{m})$. Integrating in x (Fubini):

$$\int \left[\frac{\hbar^2}{2m} |\partial_y \psi|^2 + \frac{g^2}{2} x^2 y^2 |\psi|^2 \right] \geq \int \frac{\hbar g |x|}{2\sqrt{m}} |\psi|^2,$$

and symmetrically with $x \leftrightarrow y$. Writing $2q_H(\psi) = \int \frac{\hbar^2}{2m} |\nabla \psi|^2 + \{\text{the } y\text{-slice}\} + \{\text{the } x\text{-slice}\}$ and using the two slice bounds gives $2q_H \geq q_K$ on C_0^∞ , which is Simon’s display (5) with the constants restored. The inequality extends to the form domain of H by density and shows $Q(H) \subset Q(K)$; the min-max values then satisfy $\mu_n(H) \geq \frac{1}{2}\mu_n(K)$.

Compact resolvent. Let $W \geq \gamma|x|$ be continuous with $\gamma > 0$ and $K = -\frac{\hbar^2}{2m}\Delta + W$. On the unit form-ball $\{q_K(\psi) + \|\psi\|^2 \leq 1\}$ the mass outside the ball of radius R is at most $(\gamma R)^{-1}$ and the H^1 norm on the ball is bounded, so the Rellich–Kondrachov theorem gives total boundedness in L^2 ; thus $(K+1)^{-1/2}$ and $(K+1)^{-1}$ are compact (this is the case needed of Reed–Simon IV, Theorem XIII.67). Here $W = \frac{\hbar g}{2\sqrt{m}}(|x| + |y|) \geq \frac{\hbar g}{2\sqrt{m}}\sqrt{x^2 + y^2}$. Since $\mu_n(H) \geq \frac{1}{2}\mu_n(K) \rightarrow \infty$, H has compact resolvent as well (Reed–Simon IV, Theorem XIII.16).

Simplicity. The Feynman–Kac formula $(e^{-tH/\hbar}\psi)(x) = \mathbb{E}^x[\exp(-\hbar^{-1} \int_0^t V(b_s)ds)\psi(b_t)]$ with V continuous and finite along paths shows that $e^{-tH/\hbar}$ is positivity improving for every $t > 0$, and Reed–Simon IV Theorem XIII.44 then gives a simple ground state with strictly positive eigenfunction.

Scaling. Theorem 3 with $n = 2$, $k = 4$, $\lambda = g^2/2$: $\varepsilon = \hbar^{4/3}(g^2/2)^{1/3}m^{-2/3}$, and absorbing the pure number $2^{-1/3}$ into e_n gives the displayed unit. At $g = 0$ the Fourier transform diagonalizes $-\hbar^2\Delta/2m$ as multiplication by $\hbar^2|p|^2/2m$. The classical statements are direct: on the invariant set $\{y = p_y = 0\}$ Hamilton’s equations reduce to free motion in x . $\square$

Mechanism. The valleys $\{xy = 0\}$ are unbounded and the classical phase-space volume below any positive energy is infinite, so a classical particle escapes at arbitrarily low energy. The transverse zero-point energy $\hbar\omega_x/2 = \hbar g|x|/(2\sqrt{m})$ grows linearly along a valley and confines the quantum particle. That confining term is the product of the two non-commutativities: $\hbar$ from $[x, p] = i\hbar$ and g from the quartic coupling, which is the squared commutator of the gauge potentials in Section 3.2. Removing either factor removes the gap, with the exponents $4/3$ and $2/3$ of Theorem 5(4).

3.2 The $SU(2)$ matrix model and its abelian valleys

Let $T^a = \sigma^a/2$, so $\text{tr} T^a T^b = \frac{1}{2}\delta^{ab}$ and $[T^a, T^b] = i\epsilon^{abc}T^c$; let $X_i = x_i^a T^a$ with $\vec{x}_i \in \mathbb{R}^3$, $i = 1, \dots, D$. Then $[X_i, X_j] = i(\vec{x}_i \times \vec{x}_j)^c T^c$ and $-\text{tr}[X_i, X_j]^2 = \frac{1}{2}|\vec{x}_i \times \vec{x}_j|^2$. The Yang–Mills quantum-mechanical Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} + \frac{g^2}{2} \sum_{i < j} |\vec{x}_i \times \vec{x}_j|^2 = -\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} - g^2 \sum_{i < j} \text{tr}[X_i, X_j]^2;$$

other trace or sum conventions multiply g^2 by a fixed number and every energy by its cube root.

Theorem 6. For $D \geq 2$:

1. For every ordered pair $i \neq j$, as forms, $-\frac{\hbar^2}{2m}\Delta_{\vec{x}_j} + \frac{g^2}{2}|\vec{x}_i \times \vec{x}_j|^2 \geq \frac{\hbar g}{\sqrt{m}}|\vec{x}_i|$.
2. As forms on $L^2(\mathbb{R}^{3D})$,

$$H \geq \frac{1}{2} \left[-\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} + \frac{\hbar g}{\sqrt{m}} \sum_{i=1}^D |\vec{x}_i| \right].$$

3. H has compact resolvent and a simple strictly positive ground state; its gap is $\Delta = \delta_1^{(D)} \hbar^{4/3} g^{2/3} m^{-2/3} > 0$, with the same $\hbar \rightarrow 0$ and $g \rightarrow 0$ conclusions as Theorem 5.
4. The $SO(3)$ action $(U(R)\psi)(\vec{x}_1, \dots) = \psi(R^{-1}\vec{x}_1, \dots)$ commutes with H ; the invariant subspace, which Gauss’s law selects in the constant-mode truncation of the gauge theory, reduces H , contains the ground state, and has gap $\Delta_{\text{inv}} \geq \Delta > 0$.
5. The potential vanishes exactly on the commuting configurations $\{\vec{x}_i = \lambda_i \hat{n}\}$, where all X_i lie in one Cartan subalgebra. For $D = 1$ there is no pair, $V \equiv 0$, and $H = -\hbar^2\Delta_{\mathbb{R}^3}/2m$ has spectrum $[0, \infty)$.

Proof. (1) For fixed $\vec{x}_i \neq 0$ choose an orthonormal frame of $\mathbb{R}_{\vec{x}_j}^3$ with third axis along $\vec{x}_i$. Then $|\vec{x}_i \times \vec{x}_j|^2 = |\vec{x}_i|^2(u_1^2 + u_2^2)$; drop the kinetic term in u_3 and apply the oscillator bound in u_1 and in u_2 with $\nu = g^2|\vec{x}_i|^2/2$, each giving $\hbar g|\vec{x}_i|/(2\sqrt{m})$; the complement $\vec{x}_i = 0$ is a null

set. Integrate over the remaining variables. (2) Sum the pair inequalities over all $D(D-1)$ ordered pairs with weight $w = 1/(2(D-1))$: each kinetic term T_j appears with total weight $w(D-1) = \frac{1}{2}$ and each unordered pair potential with weight $2w = 1/(D-1) \leq 1$. Then $H = \sum_{i \neq j} w(T_j + V_{ij}) + \frac{1}{2} \sum_j T_j + (1 - \frac{1}{D-1}) \sum_{i < j} V_{ij}$, where both remainders are nonnegative forms; dropping the potential remainder and using (1) gives (2). (3) The bracket in (2) has $W = \frac{\hbar g}{\sqrt{m}} \sum_i |\vec{x}_i| \geq \frac{\hbar g}{\sqrt{m}} |x|_{\mathbb{R}^{3D}}$, so the compact-resolvent, simplicity and scaling steps of Theorem 5 apply verbatim with $n = 3D$, $k = 4$. (4) $U(R)$ is unitary, preserves C_0^∞ , commutes with each $\Delta_{\vec{x}_i}$ and preserves $|\vec{x}_i \times \vec{x}_j|$; the Haar average $\int U(R) dR$ is the orthogonal projection onto the invariant subspace and commutes with $(H+1)^{-1}$, so the restriction is self-adjoint with compact resolvent. The strictly positive simple ground state ψ_0 satisfies $U(R)\psi_0 = c(R)\psi_0$ with $c(R) > 0$ and $|c(R)| = 1$, hence is invariant, so $E_0^{\text{inv}} = E_0$ and $E_1^{\text{inv}} \geq E_1$. (5) $\vec{x}_i \times \vec{x}_j = 0$ for all pairs iff all nonzero vectors are parallel. $\square$

The abelian and the two-dimensional cases. Along the commuting configurations the matrices commute and the potential is flat; writing $\vec{x}_i = \lambda_i \hat{n} + \vec{\xi}_i$ with $\vec{\xi}_i \perp \hat{n}$, the transverse curvature of the potential grows quadratically in $|\lambda|$, which is what (1) exploits. An abelian gauge group has *only* commuting configurations: its matrix model has $V \equiv 0$ and the free spectrum $[0, \infty)$. This is the exact zero-dimensional form of the statement that commutative fields have no gap, and Section 4 gives the free-field form. The temporal-gauge constant modes of a gauge theory in d spacetime dimensions are $D = d-1$ spatial components, so the zero-point mechanism needs $d \geq 3$; for $d = 2$ the single matrix has no partner, in agreement with Section 5, where two-dimensional Yang–Mills has no local excitations at all and its discrete flux spectrum comes from the compact holonomy, a different mechanism.

3.3 The finite-volume origin and the uniformity problem

Proposition 7. For $\text{SU}(2)$ Yang–Mills on a spatial torus of side L with action $S = -\frac{\hbar}{4g^2} \int F_{\mu\nu}^a F^{\mu\nu a} d^3x c dt$ (g dimensionless, $[A] = \text{length}^{-1}$), temporal gauge and the constant-mode ansatz $A_i^a(x, t) = a_i^a(t)$, the Hamiltonian is

$$H_0 = \frac{g^2 c}{2\hbar L^3} \sum_{i=1}^3 |\vec{p}_i|^2 + \frac{\hbar c L^3}{2g^2} \sum_{i < j} |\vec{a}_i \times \vec{a}_j|^2, \quad [a_i^a, p_j^b] = i\hbar \delta_{ij} \delta^{ab},$$

which is Theorem 6 with $D = 3$, $m = \hbar L^3/(g^2 c)$ and $g_B^2 = \hbar c L^3/g^2$. Its energy unit is

$$\varepsilon = \hbar^{4/3} g_B^{2/3} m^{-2/3} = \frac{g^{2/3} \hbar c}{L},$$

so the zero-mode gap $\Delta_0(L) = \delta_1^{(3)} g^{2/3} \hbar c / L$ is positive for every L and decays as $1/L$ at fixed coupling.

Proof. With $F_{0i}^a = c^{-1} \dot{a}_i^a$ and $F_{ij}^a = (\vec{a}_i \times \vec{a}_j)^a$ the Lagrangian is $\frac{\hbar L^3}{2g^2 c} \sum |\dot{\vec{a}}_i|^2 - \frac{\hbar c L^3}{4g^2} \sum_{i,j} |\vec{a}_i \times \vec{a}_j|^2$; the Legendre transform and $\frac{1}{4} \sum_{i,j} = \frac{1}{2} \sum_{i < j}$ give H_0 , and the exponent bookkeeping is $\hbar^{4/3+1/3-2/3} c^{1/3+2/3} L^{1-2} g^{-2/3+4/3}$. $\square$

Lüscher’s small-volume expansion (Nucl. Phys. B219 (1983) 233; abstract) states that the torus spectrum is discrete and expands in powers of $g^{2/3}$ with g the renormalized coupling; Proposition 7 is its leading term, and Lüscher–Münster (Nucl. Phys. B232 (1984) 445; abstract) carry it to one loop for $\text{SU}(2)$ and locate the crossover to large-volume behaviour near $z = M(0^+)L \simeq 2$. Jaffe and Witten write (p. 7 of the local PDF) that “no present ideas point the direction to establish the existence of a mass gap that is uniform in the volume”. Proposition 7 shows the difficulty in the solved sector: $\Delta_0(L) > 0$ for every L with $\inf_L \Delta_0(L) = 0$. Uniformity would need $g(L)^{2/3}/L$ bounded below, outside the weak-coupling regime where the truncation is justified; the compensation must come from the crossover regime, where the expansion no longer applies. In the language of Theorem 1, L is a fixed apparatus length that enters the monomial

and makes the finite-volume floor non-universal, and the uniformity requirement is exactly the demand that the floor be independent of it.

The supersymmetric contrast identifies the mechanism once more: de Wit, Lüscher and Nicolai (Nucl. Phys. B320 (1989) 135; abstract) prove that the supersymmetric matrix-model Hamiltonians have continuous spectrum starting at zero, through low-energy states along the same abelian valleys. The fermionic zero-point energy cancels the bosonic $\hbar\omega/2$ per transverse mode, so the confining term of Theorem 5 vanishes. The gap of Theorems 5–6 is exactly the uncanceled transverse zero-point energy.

4. Free and abelian fields have no gap; compact abelian gaps are supplied scales

Proposition 8 (free fields inherit the classical dispersion). Let H be the second quantization on Fock space of a one-particle operator $\hbar\Omega$ with $\Omega \geq 0$ the classical frequency operator. Then $\text{spec}(H) \setminus \{0\}$ has infimum $\hbar \inf(\text{spec } \Omega \setminus \{0\})$. In particular a free quantum field has a gap if and only if its classical dispersion relation has one.

Proof. On the n -particle sector H acts as $\hbar \sum_{j \leq n} \Omega_j$; its spectrum is the closure of n -fold sums of spectral values of $\hbar\Omega$, and the one-particle sector attains the infimum. $\square$

Corollary 9 (Maxwell). In a periodic box of side L , Coulomb gauge and two transverse polarizations give $H = \sum_{k \neq 0, \sigma} \hbar c |k| n_{k\sigma}$ after normal ordering, $k \in (2\pi/L)\mathbb{Z}^3$: the gap is $2\pi\hbar c/L$ and tends to zero as $L \rightarrow \infty$, where the spectrum is $[0, \infty)$. A scale-covariant classical dispersion $\omega(sk) = s\omega(k)$ has infimum zero in infinite volume; a mass term gives $\omega = \sqrt{c^2 k^2 + \mu^2 c^4/\hbar^2}$ and the supplied gap μc^2 , and gauge invariance excludes the Maxwell mass term $A_\mu A^\mu$. The constant modes are free ($k = 0$, spectrum $[0, \infty)$ already at finite L) in the non-compact theory; in compact $U(1)$ they live on a torus and contribute energies proportional to $\hbar c/L$ as well. Free quantization therefore generates no scale: every abelian finite-volume gap is $\hbar c/L$ times a number.

Proposition 10 (dimensional census of pure Yang–Mills). With $D = \partial + A$, $[A] = L^{-1}$, $[F] = L^{-2}$ and $S = \frac{1}{g^2} \int \text{tr } F^2 d^d x$ of action dimension, $[g^2] = (\text{action})^{-1} L^{d-4}$, so $g^2\hbar$ has dimension L^{d-4} . The classical field equations $D_A^* F_A = 0$ contain no constant, and the static energy of $A_s(x) = sA(sx)$ in three space dimensions is $sE(A)$, so the classical energy floor is zero. The quantum fixed constants are $\hbar$, c and g :

d	$g^2\hbar$	Mass unit from ($\hbar, c, g$)	Solved or expected gap
2	L^{-2}	$\hbar\sqrt{g^2\hbar}/c$; $g^2\hbar^2 c$ is a string tension	$E_R = \frac{1}{2}g^2\hbar^2 c L C_2(R)$ on a circle, diverging with L (Section 5)
3	L^{-1}	$g^2\hbar^2/c$	expected $m \propto g^2\hbar^2/c$ (Karabali–Nair, physics argument; lattice)
4	dimensionless	none	$m = a^{-1}F(g^2\hbar)$ with $F \rightarrow 0$ along $g(a)$: transmutation

By Theorem 1, a $d = 4$ gap positive and finite must be a multiple of a mass unit, and none exists among $(\hbar, c, g)$: the definition of the quantum theory has to introduce a scale (cutoff a or renormalization point), and the continuum limit at fixed m requires $F(g^2(a)\hbar) \rightarrow 0$ along a curve, which is dimensional transmutation stated as a dimensional necessity; the one-loop form $m \propto a^{-1} \exp[-1/(2b_0 g^2)]$ is the standard asymptotic-freedom expression (Gross–Wilczek,

Politzer 1973). In $d = 3$ the coupling itself carries the unit. In $d = 2$ its square root also permits a mass unit, while the circle spectrum below consists of electric flux. **Correction, 2026-09-26:** the earlier claim that no mass unit exists in $d = 2$ was dimensionally incorrect; absence of propagating local particles is a dynamical statement. The refinement note uses the exact two-dimensional heat-kernel construction with this distinction explicit.

Compact abelian theories. Compactness of the gauge group supplies topological excitations and, with them, gaps that are again supplied scales. Osterwalder and Seiler (Ann. Phys. 110 (1978) 440) prove reflection positivity, the strong-coupling infinite-volume limit and Wilson’s confinement bound for every compact group on the lattice; the strong-coupling mass gap follows from their cluster expansion, in units of $1/a$. Polyakov (Nucl. Phys. B120 (1977) 429) derives the photon mass of three-dimensional compact $U(1)$ from monopoles, and G pfert and Mack (Commun. Math. Phys. 82 (1982) 545) prove the mass gap and confinement of the three-dimensional Villain model for all couplings, with Debye mass $m_D^2 = (2\beta/a^3)e^{-\beta v(0)/2}$, $\beta = 4\pi^2/e^2$ in lattice units: a function of e^2a with an essential singularity that vanishes as $a \rightarrow 0$ at fixed e^2 . Guth (Phys. Rev. D 21 (1980) 2291) and Fr hlich and Spencer (Commun. Math. Phys. 83 (1982) 411) prove that four-dimensional compact $U(1)$ has a massless Coulomb phase at weak coupling. The reading levels of these citations are recorded in B78. The dimensional reading is uniform: each gapped abelian case has its gap set by an explicit scale in its definition, and the gap disappears when that scale is removed at fixed coupling; the non-abelian cases in $d = 3, 4$ differ because the coupling carries the unit or generates it by transmutation.

5. Two dimensions: an exactly solvable gap that is too large

Proposition 11. Pure Yang–Mills on a spatial circle of circumference L , in temporal gauge with Gauss’s law imposed, has as its only gauge-invariant degree of freedom the holonomy $U \in G$ up to conjugation. The physical Hilbert space is $L^2(G)^G$, the class functions, and with $S = \frac{1}{2g^2} \int \text{tr } F^2$ the Hamiltonian is $H = \frac{g^2 \hbar^2 c L}{2} C$, where C is the quadratic Casimir acting on class functions. By Peter–Weyl the eigenfunctions are the characters χ_R and

$$E_R = \frac{g^2 \hbar^2 c L}{2} C_2(R), \quad E_{\text{trivial}} = 0, \quad \text{gap} = \frac{g^2 \hbar^2 c L}{2} C_2(R_{\min}) > 0;$$

for $SU(N)$ with $\text{tr } T^a T^b = \frac{1}{2} \delta^{ab}$, $C_2(\text{fund}) = (N^2 - 1)/(2N)$, and for $U(1)$, $E_n = \frac{e^2 \hbar^2 c L}{2} n^2$ with $n \in \mathbb{Z}$ from compactness.

Proof. Gauss’s law $D_x E = 0$ makes the electric field covariantly constant along the circle, so on class functions of the holonomy E acts as the left-invariant derivative and $\int_0^L \text{tr } E^2 dx$ as L times the Laplacian on the group; $H = \frac{g^2}{2} \int \text{tr } E^2$ follows from $E = \dot{A}/g^2$, and restoring $\hbar, c$ uses $[g^2 \hbar] = L^{-2}$ from Proposition 10. Characters diagonalize the Laplacian on class functions with eigenvalue the Casimir. $\square$

The states are electric flux lines wrapping the circle, E_R/L is a string tension, and there are no propagating excitations: the theory has no local degrees of freedom. The gap grows linearly with L and diverges as $L \rightarrow \infty$, so the Jaffe–Witten finiteness requirement $m < \infty$ fails in infinite volume, where the finite-energy Hilbert space is the vacuum alone. Abelian and non-abelian groups behave identically here; the commutator plays no role in $d = 2$ because there are no transverse spatial directions, in agreement with $D = 1$ in Theorem 6. Exact solvability on the Euclidean side is Migdal’s and Witten’s (Commun. Math. Phys. 141 (1991) 153); rigorous constructions of the two-dimensional Yang–Mills measure are Driver’s, Gross–King–Sengupta’s and L vy’s (B78).

6. Transmutation without a gap: the action-critical case in mechanics

Proposition 12 (two-dimensional delta potential). For $H = -\frac{\hbar^2}{2m} \Delta - \lambda \delta^2(\vec{r})$ on $\mathbb{R}^2$, $m\lambda/\hbar^2$ is dimensionless and the classical problem has no energy scale. With a momentum cutoff $\hbar\Lambda$

the bound-state condition is

$$1 = \frac{m\lambda}{2\pi\hbar^2} \ln \left(1 + \frac{\hbar^2 \Lambda^2}{2m|E_B|} \right), \quad |E_B| = \frac{\hbar^2 \Lambda^2}{2m} \left(e^{2\pi\hbar^2/(m\lambda)} - 1 \right)^{-1}.$$

Removing the cutoff with $\lambda(\Lambda)$ chosen to keep E_B fixed defines the theory by the scale E_B alone. The spectrum is $\{E_B\} \cup [0, \infty)$.

Proof. In momentum space $\tilde{\psi}(p) = \lambda\psi(0)/(p^2/2m + |E_B|)$, and $\psi(0) = \int_{|p|<\hbar\Lambda} \tilde{\psi} d^2p/(2\pi\hbar)^2$ gives $1 = \frac{\lambda}{(2\pi\hbar)^2} \cdot 2\pi \int_0^{\hbar\Lambda} \frac{p dp}{p^2/2m + |E_B|} = \frac{m\lambda}{2\pi\hbar^2} \ln(1 + \hbar^2 \Lambda^2/2m|E_B|)$. $\square$

The scale generated by quantization sits below the continuum threshold: the excitation spectrum above the ground state is $[|E_B|, \infty)$, so the one-particle problem does have an isolated ground state with spectral gap $|E_B|$, but the “continuum” is the free particle escaping an unbounded configuration space and every excitation energy above $|E_B|$ occurs. A field-theoretic mass gap requires in addition that the vacuum be the ground state and that all excitations be massive; transmutation supplies the scale and confinement supplies the gap.

Proposition 13 (inverse-square potential). For $\ell = 0$ the radial equation of $-\frac{\hbar^2}{2m}\Delta - \lambda/r^2$ on $\mathbb{R}^3$ is $-\frac{\hbar^2}{2m}u'' - \frac{\lambda}{r^2}u = Eu$. For $2m\lambda/\hbar^2 < 1/4$ Hardy’s inequality gives $H \geq 0$, spectrum $[0, \infty)$ and no bound state. For $2m\lambda/\hbar^2 > 1/4$ the operator is not essentially self-adjoint on $C_0^\infty(\mathbb{R}^3 \setminus 0)$; each self-adjoint extension introduces a length r_0 and the bound states form the geometric tower $E_n = E_0 e^{-2\pi n/\nu}$, $\nu = \sqrt{2m\lambda/\hbar^2 - 1/4}$, accumulating at zero. Classically the fall to the centre occurs for angular momentum below $\sqrt{2m\lambda}$, an action unit: $m\lambda$ has the dimension of action squared.

This is the $k = -2$ case of Theorem 3. The classical theory owns an action unit but no energy unit; the quantum theory depends on $2m\lambda/\hbar^2$ alone, and the definition of its dynamics forces a scale. C052’s relativistic Kepler plunge at $|L| < k/c$ is a competition of the same $1/r^2$ type, so the classical action unit k/c of Q14 is the mechanical action-critical case, and C130 records its quantum version in the Dirac thresholds. The lesson for necessity arguments is the hypothesis under which “a scale is forced” is a theorem: the operator is symmetric, fails to be essentially self-adjoint, and every self-adjoint extension carries a scale. A consistency requirement, well-defined dynamics, forces the scale. Sources for Section 6: Thorn, Camblong et al., Case, Essin and Griffiths (B78).

7. The dictionary and what it decides for the positive-action question

7.1 Correspondences and non-correspondences

	Mass gap	Positive action floor
Object	physical Hamiltonian $H \geq 0$	action-defect observable on alternatives
Trivial state	vacuum Ω , $H\Omega = 0$	inertial motion, zero defect
Statement	$\text{spec } H \cap (0, m) = \emptyset$, $0 < m < \infty$	$\mathcal{A}(c) \cap (0, h) = \emptyset$, $0 < h < \infty$
Theorem 1 form	$m = (\text{number}) \times \text{mass unit of the fixed constants}$	$h = (\text{number}) \times \text{action unit of the fixed constants}$
Classical span	empty in $d = 4$	empty for (m, F) , (m, g) , (G, c)
Classical similarity	$A_s(x) = sA(sx)$	mechanical similarity of Proposition 4
Classical countermodel	continuous classical energy range	C002, C027, C056–C058

	Mass gap	Positive action floor
Free/commutative version	Proposition 8: no scale generated	ordinary calculus refinement: no scale generated
Generated gap	Theorems 5–6: two commutators, $\hbar^{4/3}g^{2/3}$	none yet; C035–C036 need a supplied reference
Supplied gap	box, lattice, mass term, $d = 2$ tension	C124/C126 (K), G05 (J), C060–C061 (speed floor)
Consistency-forced scale	regularization; Proposition 13	the target of N02
Assumed structure	gauge group, $\hbar$, quantum axioms	to be derived

The last row is the non-correspondence that governs the rest. The mass-gap problem states its non-commutative structure in the hypotheses and asks for the scale. The necessity problem of STATE item 1 asks for the structure from consistency premises, with quantum nature as the phenomenon to be explained. The analogy is therefore exact for the second stage, from structure to scale, where it says three things with proofs behind them: a positive floor is a unit multiple and needs a unit (Theorem 1); the free or commutative theory generates none (Propositions 8, Theorem 6(5)); and a generated scale needs two scale-free ingredients that fail to commute, with the gap vanishing when either is removed (Theorem 5). For the first stage, from consistency to structure, the mass-gap story contains one mechanism that does not presuppose the structure it explains: the scale forced by the requirement that the dynamics be defined at all, which is the self-adjoint-extension scale of Proposition 13 in quantum mechanics and the regularization scale of Proposition 10 in field theory.

7.2 The commutative no-floor theorem for the Galileo comparison

Proposition 14. Let the admitted class be the constant-force motions $q(t) = (vt, Ft^2/2m)$ with fixed constants m, F , all $v > 0$, $\tau > 0$ admitted, and let $X = \tau\Delta E = F^2\tau^3/(2m)$ or the area $A = vF\tau^3/(6m)$. The similarity $x \mapsto sx$, $t \mapsto s^{1/2}t$, $v \mapsto s^{1/2}v$ preserves the class and multiplies X by $s^{3/2}$ and A by s^2 ; hence both floors are zero. The dimension vectors $d_m = (1, 0, 0)$, $d_F = (1, 1, -2)$ do not span $(1, 2, -1)$, so Theorem 1 gives the same conclusion without exhibiting the similarity.

Proof. $V = Fy$ is homogeneous of degree one, so Proposition 4 with $k = 1$ gives $t \mapsto s^{1/2}t$ and action scaling $s^{3/2}$; the area is a length squared. The span statement: $a(1, 0, 0) + b(1, 1, -2) = (1, 2, -1)$ forces $b = 2$ from the length component and then $-4 \neq -1$ in time. $\square$

Which additions break the similarity? A fixed constant with action dimension, that is $\hbar$, which is circular for the necessity argument. A fixed constant whose dimension vector, with d_m and d_F , spans action: a fixed length ℓ_0 gives $\sqrt{mF\ell_0^3}$; a fixed speed c gives m^2c^3/F ; a fixed charge gives k_e/c (Q14). Each of these is a supplied unit, and the first two are mass-dependent, so they fail universality (obligation 4 of the action target) unless a further premise ties them to m . Or a structure that is itself scale-free but breaks the similarity when combined with the class: this is what the two commutators do in Theorem 5 and what the definition problem does in Proposition 13.

7.3 What the analogy transports to N02, labelled

The following is a conjecture shaped by Sections 3 and 6; it is stated so that N02 can test it, and nothing below is claimed as proved.

Conjecture (similarity anomaly). Let alternatives compose under concatenation with an action-defect observable that is additive and let the composition rule be required to be consistent in a sense to be specified (associative, positive, and defined for all admitted pairs).

If the composition rule is defined without a scale, then it is similarity-covariant and the floor is zero (Corollary 2). A positive finite floor therefore requires that the consistent definition of the composition itself carry a scale, that is, an anomaly of the classical similarity: the similarity acts on the alternatives but fails to act on the composed theory. Proposition 13 is the quantum-mechanical instance, where the composition is the self-adjoint dynamics; Proposition 10 is the field-theoretic instance, where it is the renormalized product of local operators.

What N02 would have to show is that the joint time/position refinement of the Galileo comparison poses a definition problem of this type: a composition of cut alternatives that is consistent only after a scale is chosen. The existing counterexamples are the check: C027 composes constant-force chords consistently at every mesh with no scale, and C028–C029 compose Gaussian bridges consistently with a fixed but supplied κ . A proposed principle that does not make one of these compositions ill-defined at zero scale changes nothing, exactly as a classical Yang–Mills argument that leaves $D_A^* F_A = 0$ scale-covariant proves no gap.

What the analogy cannot deliver is the first stage itself. No theorem in Sections 2–6 derives a commutator from a consistency premise; each takes $[x, p] = i\hbar$ or the gauge group as given. The claim “ $\hbar > 0$ is analogous to the mass gap” is exact and useful for the second stage and is a research heuristic, recorded here as such, for the first.

8. Strategic consequence

Gap-track work now has a written gate: name whether a proposed gap is supplied or generated, and for a generated gap name the two scale-free structures whose product produces the unit. C124/C126 and G05 are supplied gaps; G06’s perturbed cluster chain would be as well, since J remains an input. The Yang–Mills quantum-mechanical gap is the first entry in the programme whose scale is generated, and its exponents $4/3$ and $2/3$ are forced by Theorem 1 once positivity is proved; its finite-volume version exhibits the uniformity problem exactly. For STATE item 1 the consequence is a sharpened target for N02: a consistency principle that makes the zero-scale composition of cut alternatives ill-defined, tested against C027 and C028–C029. The dimensional table of Section 7.2 lists the units that a supplied-constant route would have to justify, of which only k_e/c is mass-independent. No ledger claim is promoted; Theorems 5–6 and Propositions 7, 11–13 are written proofs of established results with explicit constants, and their sources are recorded with reading levels in B78.